ZZ-B-215B
March 3, 1980
SUPERSEDING
Fed. Spec. ZZ-B-215A
August 28, 1967

FEDERAL SPECIFICATION

BELTS, V: LIGHT-DUTY OR FRACTIONAL-HORSEPOWER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers endless V-type, light-duty or fractional-horsepower drive belts.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available but are intended to cover only those generally used by the Federal Government.

1.2 Classification. The belts shall be of the following nominal cross-sectional designations, as specified (see 6.2):

Designation 2L - 1/4 inch wide by 5/32 inch thick.
Designation 3L - 3/8 inch wide by 7/32 inch thick.
Designation 4L - 1/2 inch wide by 5/16 inch thick.
Designation 5L - 21/32 inch wide by 3/8 inch thick.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 3030

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Federal Specifications:

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123	- Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-294	- Visual Inspection Guide for Rubber V-Belts.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

Military Standard Drawings:

MS39240	- Belts, V: Fractional-Horsepower (2L Cross-Sectional Designation).
MS39241	- Belts, V: Fractional-Horsepower (3L Cross-Sectional Designation).

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MS39242 - Belts, V: Fractional-Horsepower (4L Cross-Sectional Designation).
MS39243 - Belts, V: Fractional-Horsepower (5L Cross-Sectional Designation).

(Copies of Military Specifications, Standards, and Drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Mechanical Power Transmission Association
Rubber Manufacturers Association,
Engineering Standard Specification for Drives Using Multiple V-Belts 2L, 3L, 4L, and 5L Sections.

(Applications for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022, or to the Mechanical Power Transmission Association, 3525 Peterson Road, Chicago, IL 60645.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) Standards:

ANSI/ASTM D-471

Standard Test Method for Rubber Property Effect of Liquid.

(Application for copies should be addressed to American Society for Test and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The belts shall consist of tension members (inserts) and a compression member (cushion) molded into an endless loop having a trapezoidal section.

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.2.1 Rubber. Rubber shall be either natural or synthetic.

3.2.2 Fabric. Fabric shall be woven of multiple plies of natural or synthetic fibers impregnated with a rubber compound.

3.2.3 Cord. Cord shall be fabricated of natural or synthetic fibers twisted into a cord. Cords shall be prestretched and shall be impregnated with rubber compound.

3.2.4 Tension members. Tension members shall be either fabric or cord reinforcement and shall be arranged in a symetrical pattern parallel to the circumference of the belt.

3.3 Construction.

3.3.1 Spacing of members. Tension members shall be uniformly spaced in a direction parallel to the direction of the load and imbedded in the belt structure.

3.3.2 Splicing. Not more than two transverse splices shall be allowed in any ply of fabric in any belt having a length of 65 inches or less. Belts over 65 inches shall have not more than three transverse splices in any ply of fabric.

3.3.3 Notching or cogging. The inner circumference of the belt may be plain, notched, or cogged.

3.3.4 Covers. Covers, if used, shall consist of one or more plies of stretchable fabric or bias cut cord, frictioned or impregnated for adhesion.

3.3.5 Compression member. The compression member shall be of natural or synthetic rubber compounds and may include natural or synthetic fiber cords or fabric.

3.4 Dimensions. Nominal cross sectional dimensions, effective outside lengths, length tolerance, and belt ride and bottom clearance (see figure 1) shall be as specified in MS39240 through MS39243, as applicable.

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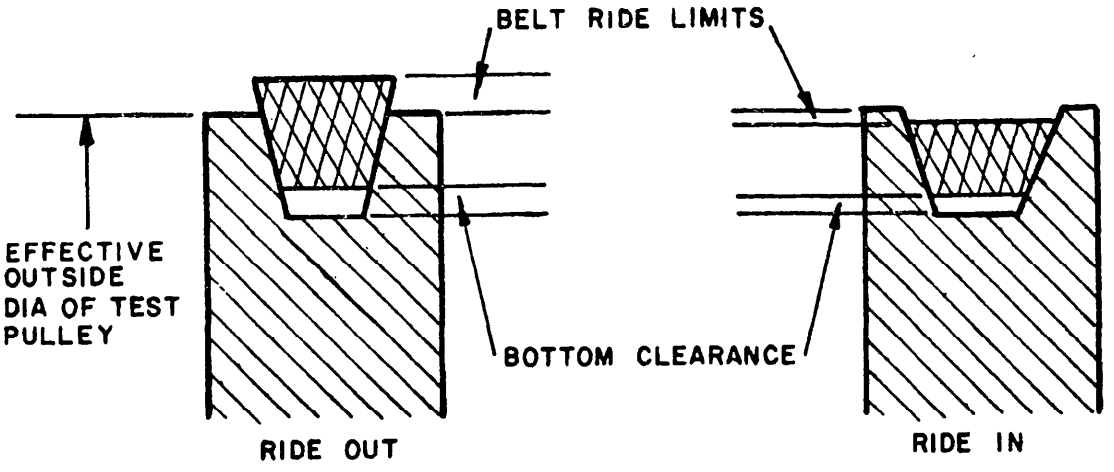


FIGURE 1. Belt ride and bottom clearance.

X-3502

3.5 Matched sets. When specified (see 6.2) belts shall be furnished in matched sets according to designation, length, and quantity specified. Each belt in a matched set shall conform to the requirements for the single belt and the difference in effective outside lengths of belts in a matched set shall not exceed 1/10 inch.

3.6 Identification marking. The belts shall be marked on the outside circumference in accordance with MIL-STD-130. The identifying number shall be the industry number in accordance with MS39240 through MS39243, as applicable and the manufacturer's name or trademark. When age is specified (see 6.2), the marking shall include the quarter and year (such as 1Q79) of manufacture. Ink imprinting or molded markings are acceptable.

3.6.1 Matched sets. Each belt of a matched set shall be marked as specified in 3.6. In addition, all belts of a matched set shall be tied together and include a tag or label printed with the following notice: "Warning, this is a matched set. Do not break ties except at installation."

3.7 Age. When specified (see 6.2), the belts shall be not more than 12 months old (4 calendar quarters) when submitted for acceptance (see 6.3).

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3.8 Electrical resistivity. Electrical resistance shall not be greater than 6 megohms when tested as specified in 4.4.2.3.

3.9 Oil-Resistance. When specified (see 6.2), the belt thickness shall increase no more than 15 percent after exposure in ASTM No. 3 Oil, and shall decrease (shrink) no more than 1 percent after exposure in ASTM No. 1 Oil, when tested as specified in 4.4.2.4.

3.10 Resistance to temperatures. The belts shall not crack or require a greater torque to start or continue rotation than as shown in table I, when tested, as specified in 4.4.2.5.

TABLE I. Maximum Torque Requirements

Cross-sectional designation	Torque required to start maximum (foot pounds)	Torque required to continue maximum (foot pounds)
2L	20	12
3L	26	14
4L	28	15
5L	30	18

3.11 Workmanship. The belts shall be free of any visible defects listed in MIL-STD-294.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-3.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects. Major and minor defects are defined in MIL-STD-294.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1 and 4.4.2.2. AQL shall be 2.5 percent defective. A belt shall be determined defective upon failure to pass any test.

4.4 Inspection procedure.

4.4.1 Examination. The belts shall be examined for the defects listed in MIL-STD-294.

4.4.2 Tests.

4.4.2.1 Effective outside length. The apparatus for this test shall consist of two equal-diameter pulleys, one which is stationary and the other one shall be moveable along a graduated scale (see figure 2). The pulley dimensions and tension shall conform to the values shown in table II. Place the belt in the test pulleys and apply the total tension, then rotate the belt around the pulleys not less than two complete revolutions to seat the belt in the pulley grooves and to equally divide the tension between the two strands. The effective outside length of the belt is obtained by adding the effective outside circumference of one of the measuring pulleys to twice the center distance between the two pulleys. Nonconformance to length tolerances specified in MS39240 through MS39243, as applicable, or the requirements for matched sets as specified in 3.4, shall constitute failure of this test.

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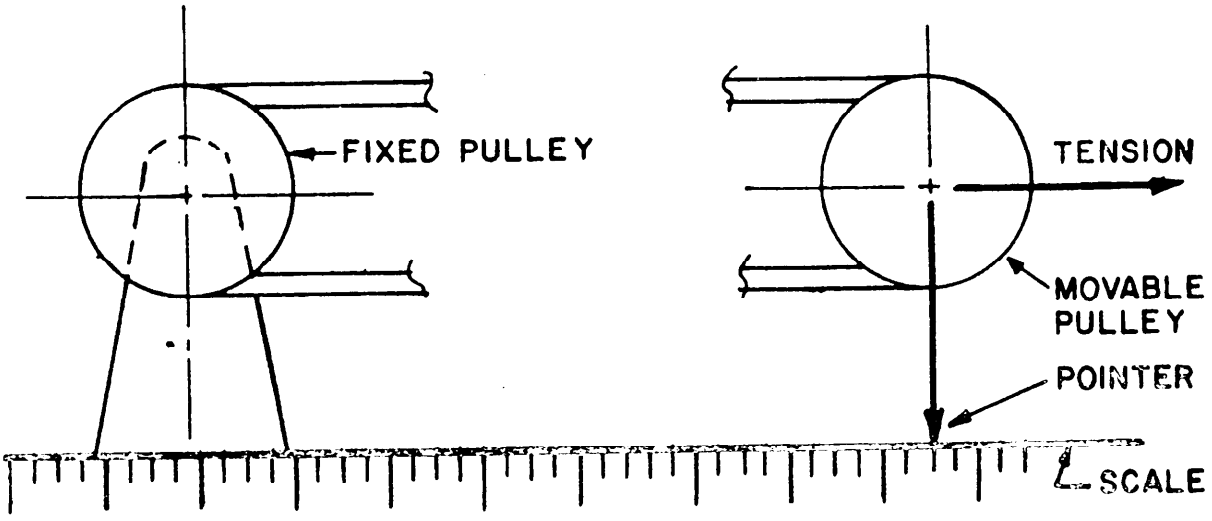


FIGURE 2. Apparatus. X-3503

Table II. Inspection Pulley Sizes

Design-nation	Effective outside diameter + 0.001	Effective outside circumference	Groove angle + 10 min.	Groove top width + 0.001	Minimum groove depth	Tension (Total) 1/
	Inches	Inches	Degrees	Inch	Inch	Pounds
2L	1.273	4.000	32	0.240	0.250	10
3L	2.546	8.000	34	0.364	0.406	25
4L	3.183	10.000	32	0.490	0.490	35
5L	4.775	15.000	32	0.630	0.580	50

1/ Tension on each strand is one-half the amount shown.

4.4.2.2 Belt ride and bottom clearance. Following the length determination test, with the applicable tension and rotation as specified in 4.4.2.1, the belt ride and bottom clearance shall be determined (see figure 1). The belt ride shall be measured from the top of the belt to the effective outside diameter of the pulley. Bottom clearance shall be computed by subtracting the actual overall thickness of the belt from the depth of the pulley groove, and to the remainder add or subtract the belt ride measurement. A belt ride or bottom clearance exceeding the limits specified in the applicable military standard shall constitute failure of this test.

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4.4.2.3 Electrical resistivity. Place two brass contacts on the belt specimen 8-1/2 inches on centers. Each contact shall be moistened with water and pressed against the belt specimen with a force of 12-1/2 pounds. Resistance between contacts shall be measured with an ohmmeter operating at a potential of 500 volts. The ohmmeter shall have a range from zero to 10 megohms with an accuracy of plus or minus 2 percent. A resistivity greater than 6 megohms between contacts shall constitute failure of this test.

4.4.2.4 Oil-resistance test. Three belt sections each, 3 inches long, shall be immersed in ASTM Oil No. 1 and ASTM Oil No. 3, as cited in ASTM D-471, at a temperature of 158° for 22 hours. The thickness of each belt section shall be measured at the center and 1/2-inch from each end, before and after the immersion. The percent increase or decrease in thickness of any specimen shall be recorded as the average of the three readings for that specimen. Failure of any specimen to meet the requirements of 3.9 shall be cause for rejection.

4.4.2.5 Temperature test. The belts shall be placed in the air oven in an unrestrained, horizontal position and the temperature of the oven shall be maintained at 180° ± 5° F. for 12 hours. At the end of 12 hours, the belt shall be removed from the oven and allowed to cool for 30 minutes at 75° ± 10° F. Then the belt shall be installed on the pulley and panel assembly and the belt tensioned by applying the applicable tension weight specified in table II. The tension pulley location shall be locked and the weight removed. The assembly of belt and pulleys shall be placed in the cold chamber and the chamber maintained at a temperature of minus 40° ± 5° F for 12 hours. At the end of 12 hours and while still in the cold chamber, the pulleys shall be rotated with a torque wrench applied to the driver pulley until the belt has been rotated not less than two complete revolutions. The torque required to start and the torque required to continue rotation shall be recorded. The belt shall then be examined for visible evidence of cracking. Any evidence of cracks or torque in excess of that specified in table I shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101. Materials, or containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
102. Matched belt sets not preserved as single unit.
103. Belts not preserved in quantities specified.
104. Gross weight exceeds the weight limitation of the fiberboard box for level B.
105. Strapping not as specified.
106. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B or commercial as specified (see 6.2).

5.1.1 Level A. Each matched belt set shall be preserved as a single unit (see 3.6.1). Belts of like description or belt sets of like description shall be preserved together in the quantities specified (see 6.2) in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the box specification appendix, method V.

5.1.2 Level B. Each matched belt set shall be preserved as a single unit (see 3.6.1). Belts of like description or belt sets of like description shall be preserved together in the quantities specified (see 6.2), in a close-fitting box conforming to PPP-B-636, W6c, style optional.

5.1.3 Commercial. Each matched belt set shall be preserved as a single unit. Belts and belt sets shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. The belts or belt sets of like description shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B.

5.2.2.1 Military agencies. The belts or belt sets of like description packaged as specified in 5.1 shall be packed as specified for level A except domestic boxes may be used or shall be packed in close-fitting boxes

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conforming to PPP-B-636, V3c or V3s. The gross weight shall not exceed the weight limitation of the box specification. Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2.2 Civil agencies. The belts or belt sets of like description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The gross weight shall not exceed the weight limitation of the box specification. Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. The belts or belt sets of like description shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military packaging. Shipments shall be marked in accordance with MIL-STD-129.

5.3.3 Commercial packaging. Shipments shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Belts covered by this specification are intended for use with fractional-horsepower engines and motors, using small-diameter pulleys on drives which generally have requirements within the capacity of a single belt. Light-duty or fractional-horsepower belts are also intended for intermittent or infrequent, rather than continuous operation. Typical applications are domestic washing machines, small fans and blowers, centrifugal pumps, garage equipment, and workshop equipment.

6.1.1 2L designation. The 2L designation is in limited usage and is not manufactured by all V-belt manufacturers.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Nominal cross-sectional designation and lengths required (see 1.2 and 3.4).
(c) When matched sets are to be furnished; also the cross-sectional designation, length, and quantity of belts per set (see 3.5).
(d) When age of belts shall be not more than 12 months old (4 calendar quarters) (see 3.7).

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(e) When oil-resistance is required (see 3.9).
(f) Degree of preservation and packing required (see 5.1 and 5.2).
(g) Quantity of belts per package (see 5.1.1 and 5.1.2).
(h) When finish A strapping is required (see 5.2.1).

6.3 Age. The age requirement shall be imposed only for large quantity procurement of belts that are intended for replacement purposes over a period of several years.

MILITARY INTEREST:

Custodians:
Army - ME
Air Force - 99

Review activities:
DLA - CS
Other - NSA

User activities:
Army - AV

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS
DOT - ACO

Preparing activity:
Army - ME

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